$$\Rightarrow \quad \frac{|z_2z_3 + z_1z_3 + z_1z_2|}{|z_1 + z_2 + z_3|} = r. \qquad \text{(given that } z_1 + z_2 + z_3 \neq 0\text{)}$$

Thus, $$\left|\frac{z_2z_3 + z_1z_3 + z_1z_2}{z_1 + z_2 + z_3}\right| = r.$$

Example 2.16

Show that the equation $z^2 = \overline{z}$ has four solutions.

Solution

We have, $$z^2 = \overline{z}.$$

$$\Rightarrow |z|^2 = |z|$$

$$\Rightarrow |z|(|z| - 1) = 0,$$

$$\Rightarrow |z| = 0, \text{ or } |z| = 1.$$

$|z| = 0 \Rightarrow z = 0$ is a solution, $|z| = 1 \Rightarrow z\overline{z} = 1 \Rightarrow \overline{z} = \frac{1}{z}$.

Given $z^2 = \overline{z} \Rightarrow z^2 = \frac{1}{z} \Rightarrow z^3 = 1.$

It has 3 non-zero solutions. Hence including zero solution, there are four solutions.

2.5.2 Square roots of a complex number

Let the square root of $a + ib$ be $x + iy$

That is $\sqrt{a+ib} = x + iy$ where $x, y \in \mathbb{R}$

$$a + ib = (x + iy)^2 = x^2 - y^2 + i2xy$$

Equating real and imaginary parts, we get

$$x^2 - y^2 = a \text{ and } 2xy = b$$

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2 = a^2 + b^2$$

$$x^2 + y^2 = \sqrt{a^2 + b^2}, \text{ since } x^2 + y^2 \text{ is positive}$$

Solving $x^2 - y^2 = a$ and $x^2 + y^2 = \sqrt{a^2 + b^2}$, we get

$$x = \pm\sqrt{\frac{\sqrt{a^2 + b^2} + a}{2}} \; ; \; y = \pm\sqrt{\frac{\sqrt{a^2 + b^2} - a}{2}}.$$

Since $2xy = b$ it is clear that both x and y will have the same sign when b is positive, and x and y have different signs when b is negative.

Therefore $\sqrt{a+ib} = \pm\left(\sqrt{\frac{|z|+a}{2}} + i\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right)$, where $b \neq 0$. $\left(\because \text{Re}(z) \leq |z|\right)$

Formula for finding square root of a complex number

$\sqrt{a+ib} = \pm\left(\sqrt{\frac{|z|+a}{2}} + i\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right)$, where $z = a + ib$ and $b \neq 0$.

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